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Roll no.

Degree: B-Tech, Semester: 2nd
MID-SEMESTER EXAMINATION, February 2024

Course Title: Digital Circuits & Systems
Course Code: INECC01

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Convert the binary sequence $(1011001)_2$ into Gray code	1.5	CO1
1b	Perform the addition $(34)_{10}$ and $(25)_{10}$ in BCD	1.5	CO1
2a	Implement the logic function $F(A,B,C) = \sum m(2,4,6,8)$ using 3x8 decoder and logic gates	1.5	CO1
2b	Draw internal gate level circuit diagram for a 4x1 MUX	1.5	CO1
3a	Implement the logic function $F(A,B,C) = AB+BC+CA$ using 2-input NAND gates only	1.5	CO1
3b	Minimize the logical expression represented by the F $(x,y,z) = \pi M(2,5,6,7)$ and realize using logic gates	1.5	CO1
4a	Derive the expression for difference and borrow in a full subtractor	1.5	CO1
4b	Design a combinational circuit which produces square of a 3-bit input binary number	1.5	CO1
5a	Explain why NAND and NOR gates are called universal logic gates	1.5	CO1
5b	Differentiate between active high and active low signals	1.5	CO1